Vision Encodes, Language Routes: Token-Level Functional Specialization in a Vision-Language-Action Model

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Abstract: Vision-language-action models (VLAs) fine-tune large language models (LLMs) on robot demonstrations, but what fine-tuning changes at the representational level remains unclear. We present a systematic framework for token-level analysis of VLAs, applying cross-instruction probes, attention analysis, Representational Similarity Analysis (RSA), and functional probes across all 32 LLM layers of OpenVLA-7B on LIBERO, with factorial trained/pretrained/untrained controls. Three findings emerge. First, VLA fine-tuning front-loads instruction routing: language pretraining concentrates instruction attention at mid-depth, while VLA training shifts this to entry-layer routing (L0–L1) with exit-layer reengagement (L31). Seven complementary analyses, including dose-response ablation (8 conditions, monotonic $d = 0.48 \rightarrow 0.64$) and counterfactual instruction analysis (4 perturbation types, $d = 0.63$–0.97), support instruction-conditioned action generation through distributed L1 routing, with routing dependence $2.5\times$ stronger for instruction-discriminative tasks ($p = 0.033$). Second, vision encoder representations show a metric-dependent middle-layer peak (RSA/probes corroborate; CKA diverges), with training effects concentrated at the readout token (R^2 $+0.069$, attention $+46$pp) rather than visual geometry. Architecture-expected token specialization (instruction-blind image tokens, perfect readout classification from L1) is empirically verified as a framework sanity check.

1 Introduction

What does fine-tuning change inside a Vision-Language-Action model (VLA)? VLAs fine-tune large language models on robot demonstrations to produce actions from visual observations and language instructions [1, 2], but the decoder-only causal mask prevents image tokens from attending to instructions. Instruction information must reach action-generating positions through learned attention routing, yet how fine-tuning reshapes this routing remains uncharacterized.

Recent behavioral studies reveal that VLAs rely disproportionately on visual input [3], with language instructions largely ignored by the action pathway [4]. These findings suggest internal specialization but do not explain *how* instruction information reaches the action-generating readout token, *what* routing strategy fine-tuning learns compared to language pretraining alone, or *whether* training creates new representational structure or reconfigures inherited geometry. Existing analyses have not combined systematic per-layer probing with factorial trained/pretrained/untrained controls, leaving the representational mechanisms of VLA fine-tuning opaque.

We address this gap by decomposing OpenVLA-7B [2, 5], a VLA chaining a frozen DinoSigLIP vision backbone [6, 7], an MLP projector, and Llama-2 7B [8]. We extract representations from each of 32 LLM layers at both image-token and readout-token positions, and from external baselines (DINOv2 [7], CLIP [9]). We compare three LLM conditions (VLA-trained, pretrained Llama-2 base,

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and randomly initialized) crossed with instruction presence, and apply four complementary analyses: cross-instruction probes for token-level specialization, L1 attention pattern analysis for routing mechanisms, RSA [10] for geometric alignment, and MLP probes [11] for functional decodability.

Three findings emerge from our analysis of OpenVLA-7B on LIBERO. *First*, VLA fine-tuning front-loads instruction routing: language pretraining concentrates instruction attention at mid-depth (L9–L16, 15–22 heads), while VLA training shifts this to an entry-exit pattern (L0–L1: 21–24 heads, L31: 18 heads). Seven complementary analyses, including dose-response ablation (8 conditions, monotonic $d = 0.48 \rightarrow 0.64$) and counterfactual instruction analysis (4 perturbation types, $d = 0.63$–0.97), support instruction-conditioned action generation through distributed L1 routing, with routing dependence $2.5\times$ stronger for instruction-discriminative tasks ($p = 0.033$). *Second*, under RSA and action-probe decodability, vision encoder representations show a middle-layer peak (R^2 at L8–L14, RSA at L12), but this pattern is metric-dependent (CKA diverges; Appendix A.1). Training yields substantial functional improvement at the readout (R^2 $\Delta = +0.069$, $p < 0.0001$) with limited geometric gain ($\Delta\rho = 0.017$ on LIBERO-Goal; significant on LIBERO-Spatial, $p < 0.001$). Architecture-expected token specialization (instruction-blind image tokens, perfect readout classification from L1) is empirically verified as a framework sanity check (§4.2).

Our contributions are: (i) evidence from seven complementary analyses that VLA fine-tuning front-loads instruction routing from mid-depth to entry-exit, with L1 as the primary routing hub and routing dependence modulated by task instruction-discriminability; (ii) the observation that vision encoder representations follow a metric-dependent middle-layer peak (RSA/probes corroborate; CKA diverges), with training primarily enhancing readout-token integration; and (iii) a systematic framework for token-level representational analysis of VLAs with factorial controls.

2 Related Work

Vision-language-action models. VLAs fine-tune large language models on robot demonstrations to produce action tokens conditioned on visual observations and language instructions [1, 2, 5, 12]. This architecture family inherits the transformer backbone and its multi-modal fusion mechanisms from vision-language models, but repurposes them for continuous control. A central assumption is that fine-tuning injects action-relevant knowledge into pretrained representations. We test this assumption at the representational level through per-layer, per-token-type decomposition.

Mechanistic interpretability in transformers. A growing body of work analyzes transformer internals at the level of individual attention heads and circuits. Elhage et al. [13] introduced the residual stream framework, in which each layer reads from and writes to a shared communication channel. Building on this, Olsson et al. [14] identified induction heads as a mechanistic basis for in-context learning; these specialized attention heads implement token-copying algorithms. Circuit-level analyses have since traced complete computational subgraphs for tasks such as indirect object identification [15] and greater-than comparison [16]. Meng et al. [17] developed causal tracing for factual associations, Belrose et al. [18] introduced the tuned lens for per-layer prediction, and feature-level methods have scaled to dictionary learning over millions of features [19, 20]. Attention probing has shown that individual heads encode syntactic relations [21], capture depth-dependent patterns [22], and that a small subset accounts for most performance [23]. Our Layer-1 attention head analysis extends this tradition to the VLA domain; unlike knowledge editing or prediction lens approaches, we use functional ablation to study task-specific routing and find that 24 of 32 heads become instruction-dominant after fine-tuning.

Behavioral and mechanistic analysis of VLAs. Behavioral evaluations reveal that current VLAs rely disproportionately on visual input, with language instructions largely ignored during action generation [3, 24, 25]. At the mechanistic level, Häon et al. [4] identified sparse semantic directions in VLA residual streams linked to action selection. Sparse autoencoder studies have uncovered interpretable features and memorized action sequences [26, 27], while probing work has tested for symbolic state representations and emergent world models [28, 29]. Fang et al. [30] showed that fine-

tuning erodes general VLM reasoning capabilities. These studies analyze individual components or aggregate statistics; we provide a systematic per-layer decomposition with trained-vs-untrained controls absent from prior work. The closest work to ours is Grant et al. [3], who performed a feature-level mechanistic study of VLAs and found visual dominance in internal representations. Our work differs in three respects: (1) we decompose at the token level rather than the feature level and identify the readout token as the sole multimodal integration site; (2) we track representations across all 32 layers and reveal a non-monotonic transformation profile invisible to whole-model analyses; and (3) we include trained-vs-untrained factorial controls that isolate the effect of fine-tuning on attention routing mechanisms.

Token-level functional specialization in transformers. In Vision Transformers, the CLS token aggregates global information [31] and certain patch positions serve as registers [32]; in language models, feed-forward layers store factual associations in position-specific patterns [33], probing reveals increasingly abstract features across depth [34], the logit lens [35] tracks prediction evolution, and Neo et al. [36] identified task-specific neurons in multi-task LLMs. We extend this to VLAs: image tokens remain instruction-blind while the readout token serves as the exclusive multimodal hub.

Representation similarity and probing methodology. RSA [10], SVCCA [37], CCA [38], and Centered Kernel Alignment (CKA) [39] compare geometric structure between representation spaces; Raghu et al. [40] applied these tools to compare ViT and CNN representations. Linear probes [11, 41, 42, 43] complement geometric methods with functional evaluation. We adopt RSA and MLP probes, with CKA in the appendix.

3 Method

We investigate how action-relevant structure is distributed across a VLA by combining four complementary analyses: cross-instruction probes, attention pattern analysis, RSA, and functional probes.

3.1 Model and Data

We study OpenVLA-7B [2, 5], chaining a frozen DinoSigLIP vision backbone [6, 7] (DINOv2 ViT-L/14 + SigLIP-SO400M), an MLP projector (2,176-D $\rightarrow$ 4,096-D), and Llama-2 7B [8] (6.74B parameters) that autoregressively decodes 7-D actions. We use the base checkpoint pretrained on Open X-Embodiment [44] (970K episodes spanning multiple robot platforms and task types) and evaluate on LIBERO-Goal [45] (10 tasks, 50 demonstrations each), which was not part of the pretraining mixture. We compare three LLM conditions: (1) VLA-trained (released checkpoint), (2) Llama-2 base (pretrained backbone, no VLA fine-tuning), and (3) untrained (6.74B parameters reinitialized via `from_config()`); a fourth condition (task-specific fine-tuning on LIBERO) is compared in §4.3. All three share the same frozen vision encoder and projector.

3.2 Representation Extraction

We extract representations from six sources spanning the VLA pipeline: (a) **DinoSigLIP+Projector**, the projector output (4,096-D), immediately before LLM processing; (b) **LLM image_mean**, the mean image-token activation at each of 32 layers; (c) **LLM readout**, the hidden state at the final pre-action token position at each layer (located after both image and instruction tokens in the causal sequence, it can attend to all preceding context and is thus the natural site for multimodal integration before action prediction); (d) **DINOv2 ViT-L/14** [7] (`cls_token/patch_mean`), a self-supervised baseline; (e) **CLIP ViT-L/14** [9] (`patch_mean`), a VL-pretrained baseline; (f) **Branch-split**, the DINOv2 (1,024-D) and SigLIP (1,152-D) branches before fusion. Throughout, L0 denotes the embedding layer output, L1–L31 the 32 transformer layers, and L32 the output after the final layer norm. We cross two factors: *training status* (trained vs. untrained) and *instruction presence* (with vs. without task description) in a 2×2 factorial design.

3.3 Cross-Instruction Probes

To test whether different token types encode instruction identity, we construct a cross-instruction dataset: each visual frame is paired with all 10 task instructions, producing 10 representation vectors per frame per layer. A linear classifier (10-way) is trained to identify which instruction was present, separately for image_mean and readout token positions. We validate with leave-one-out cross-validation (LOOCV) over 10 folds, each holding out one task's instructions, to verify that results reflect generalizable instruction encoding rather than memorization.

3.4 Attention Pattern Analysis

We analyze attention distributions at the readout token position across all 32 layers, categorizing each head as instruction-dominant ($> 50\%$ attention to instruction tokens) or image-dominant, with detailed per-head decomposition at L1. This complements the cross-instruction probe by revealing the *mechanism* through which instruction information reaches the readout token.

3.5 Representational Similarity Analysis

We adopt RSA [10] to quantify geometric alignment between representations and the action space. For each task, we compute pairwise $N \times N$ representational dissimilarity matrices (RDMs) using cosine distances for representations and Euclidean distances for 7-D action vectors, using within-task observation pairs from 50 demonstrations. Alignment is measured via Spearman ρ between the upper-triangular entries of the representation and action RDMs; higher ρ indicates that representationally similar observations require similar actions. We report mean ρ across 10 tasks with 95% confidence intervals from 10,000 bootstrap resamples; resampling is performed at the task level ($N = 10$), treating each task's pairwise RDM entries as a block to preserve within-task dependency structure [46].

3.6 Functional Action Probes

RSA measures geometric alignment but not decodability. We complement it with MLP probes [11]: two-layer networks ($d_{\text{in}} \to 256 \to 7$) trained to predict 7-D action vectors from frozen representations (Adam, lr $= 10^{-3}$, batch 256, 50 epochs, early stopping patience 5), with demo-level train/test splits (demos 0–39 vs. 40–49). We report per-task macro-averaged R^2 across 10 tasks with bootstrap paired comparisons (10,000 resamples).

Why RSA over CKA. RSA captures relative distance structure via scale-invariant Spearman rank correlation, suited to comparing high- and low-dimensional spaces (4,096-D vs. 7-D). CKA [39] measures global subspace overlap, which can remain high even when geometric structure differs. We report both in Appendix A.1; the divergence between metrics (§4.4) indicates neither alone fully characterizes representational structure.

4 Experiments

4.1 Experimental Setup

We evaluate OpenVLA-7B [2, 5] on LIBERO-Goal [45] (10 tasks, 50 demonstrations each). Representations are extracted from six sources spanning the VLA pipeline (§3.2): DinoSigLIP+Projector, LLM image_mean and readout tokens at each of 32 layers, DINOv2 [7] (`cls_token`/`patch_mean`), CLIP [9] (`patch_mean`), and branch-split (DINOv2 1,024-D / SigLIP 1,152-D before fusion). Three LLM conditions are compared: VLA-trained, pretrained Llama-2 base (no VLA fine-tuning), and untrained (6.74B parameters reinitialized to random weights), crossed with instruction presence (§3.1). RSA computes Spearman ρ between cosine-distance representation RDMs and Euclidean-distance action RDMs, with 10,000 bootstrap CIs (§3.5). MLP probes (input $\to 256 \to 7$; Adam, lr $= 10^{-3}$, 50 epochs, early stopping) report mean R^2 (§3.6). This yields $N = 63{,}228$ observations (after

Table 1: Cross-instruction 10-way classification at selected layers (full profile in Appendix A.18.1). Image_mean remains at chance (10%) in all conditions; trained readout reaches 100% at L1 and holds; untrained readout decays from L1 to L31.

Layer	Trained readout	Untrained readout	Image_mean
L0	10.0%	10.0%	10.0%
L1	100.0%	100.0%	10.0%
L4	100.0%	99.1%	10.0%
L8	100.0%	98.2%	10.0%
L16	100.0%	92.3%	10.0%
L31	100.0%	82.0%	10.0%

excluding terminal frames without next-action labels; 63,728 total); RSA uses task-level blocking ($N_{\text{eff}} = 10$) and probes use demo-level train/test splits. All patterns reproduce with a linear probe (Appendix A.10).

4.2 Framework Validation: Token-Level Specialization

As a sanity check on the experimental framework, we train cross-instruction probes (§3.3): given the same visual frame with each of 10 task instructions, we classify which instruction was present from per-layer activations.

Image tokens are instruction-blind (expected). As predicted by the causal mask and frozen encoder, image_mean instruction classification remains at chance (10%) across all 32 layers in all conditions (Table 1). Both linear and MLP probes corroborate this, as does standalone DINOv2 (10%; Appendix A.12).

Readout token as multimodal hub. Trained readout achieves 100% 10-way classification from L1 through L31 (LOOCV-confirmed; Appendix A.18.2); untrained achieves 100% at L1 but decays to 82% by L31. The readout is thus the sole locus of multimodal integration, with richer task-relevant content than standalone DINOv2 (QA probe 86.3% vs. 82.7%; Appendix A.18.3), replicating on LIBERO-Spatial (Appendix A.14).

4.3 VLA Fine-Tuning Front-Loads Instruction Routing

Three-level routing gradient at L1. L1 readout attention across three conditions (Table 2; Figure 1): untrained allocates 0/32 instruction-dominant heads (11.7% text); Llama-2 base 9/32 (36.5%); VLA fine-tuning 24/32 (57.7%), a redistribution beyond inherited priors. Per-token normalization (38.2× text-to-image ratio; Appendix A.19) corroborates this redistribution.

Redistribution of routing across depth. The 32-layer profile (Figure 1) shows qualitatively distinct patterns. Llama-2 base concentrates instruction attention at mid-depth (L9–L16: 15–22 heads). VLA fine-tuning shifts to an entry-exit pattern (L0–L1: 21–24 heads, L31: 18), with intermediate layers markedly lower (0–11 at L2–L30); ordering is stable across thresholds (30%–70%; Appendix A.18.7).

Task-specific sharpening. A fourth condition (OpenVLA-7B fine-tuned on LIBERO) shows stable L1 entry routing (22/32 heads) but complete loss of L31 exit re-engagement (0 vs. 18 heads) and further mid-depth compression (L9–L16: 1–7 vs. 6–11), sharpening entry-exit routing to entry-only.

Probe robustness corroborates the pattern. Cross-instruction probes (Appendix Figure 14) corroborate: VLA-trained achieves 99.6% at L1; Llama-2 base starts at 80.2%, saturating by L5; untrained decays from 100% to 82% (L31). Training effects concentrate at the readout (R^2 +0.069, attention +46pp) while image_mean changes minimally ($\Delta \leq 1.7\%$; Appendix A.17, A.6).

Table 2: Instruction routing at L1. Mode: immed./gradual/incid. = acquisition speed; +stable/+forget = retention. Accuracy gap (99.6% vs. 100.0% in Table 1): task-level vs. LOOCV splits.

Condition	Heads	Text attn	L1 acc	L31 acc	Mode
VLA Trained	24/32	57.7%	99.6%	100.0%	immed.+stable
Llama-2 base	9/32	36.5%	80.2%	100.0%	gradual+stable
Untrained	0/32	11.7%	100.0%	82.0%	incid.+forget

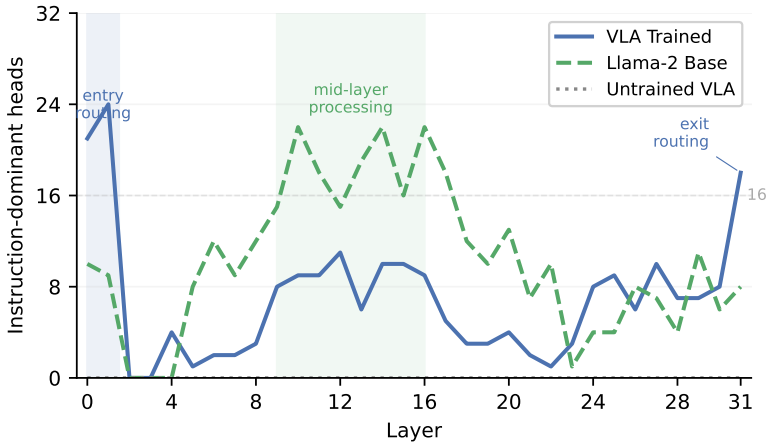


Figure 1: Instruction-dominant heads ($> 50\%$ text) across 32 layers. VLA-trained: entry-exit (L0–L1: 21–24, L31: 18); Llama-2 base: mid-depth (L9–L16: 15–22); untrained: zero. Probe accuracy in Appendix Figure 14.

Open-loop functional sensitivity. We ablate attention to instruction tokens at the readout position. Ablating all 24 dominant heads at L1 increases action R^2 by $+0.042$ (baseline 0.517)[1]. We compare against 10 random subsets of 24 heads (Appendix A.7): random ablation yields a larger mean increase of $+0.077 \pm 0.013$ ($z = -2.73$, $p = 0.003$). Dominant heads produce a smaller gain because they carry organized routing signals (structured noise for the probe); random heads lack such organization. In the original extraction pipeline (§3.2), L1 ablation increases R^2 by $+0.059$ vs. no-ablation baseline, decaying at deeper layers (L8: $+0.018$; L13: $\Delta \approx 0$); instruction routing is an entry-layer phenomenon, consistent across all 10 tasks (Table 20).

Action trajectory validation. Ablating 24 instruction-dominant heads alters 72.3% of action tokens ($d = 0.65$); control ablation (8 heads) yields $d = 0.41$ (29.3%), concentrated on the gripper dimension (shift 0.291 vs. < 0.034; Tables 18, 19; Appendix A.8).

Dose-response gradient. Ablating 4–32 instruction-dominant heads yields monotonic divergence ($d = 0.482 \rightarrow 0.639$; Figure 8), with steep rise through 12 heads and plateau beyond 20 (Appendix A.8).

Counterfactual instruction analysis. Replacing the original instruction with four perturbation types all produce substantial divergence: empty ($d = 0.968$), wrong task (0.844), paraphrased (0.697), shuffled (0.625; Table 6). Token novelty ($1 -$ Jaccard) predicts the ordering ($\rho = 1.0$): paraphrased (Jaccard 0.533) exceeds shuffled (Jaccard 1.0), though shuffled $d = 0.625$ with identical tokens reveals positional sensitivity. L1 ablation ($d = 0.65$) falls within this range (Appendix A.8).

[1] R^2 values differ from Table 3 due to the feature reconstruction pipeline; all comparisons are within-experiment.

Table 3: RSA ρ and probe R^2 alignment with actions. DINOv2 achieves highest RSA (0.692); ordering is metric-dependent (Appendix A.1). VLA rows: peak values. **Bold**: best; underline: second.

Source	RSA ρ	R^2	RSA Peak	R^2 Peak
DINOv2 cls_token	**0.692**	0.634	—	—
VLA Trained readout	0.643	**0.678**	L12	L13
Fused DinoSigLIP	0.634	—	—	—
VLA Trained image_mean	0.628	0.655	L12	L8–L10[†]
DinoSigLIP+Projector	0.615	0.630	—	—
VLA Untrained image_mean	0.612	0.640	L0	L0
VLA Untrained readout	0.626	0.609	L1	L32
CLIP ViT-L/14[‡]	0.557	0.587	—	—

[†] R^2 peak region across 5 seeds (0.655 ± 0.002; Appendix A.15.4).
[‡] RSA uses `patch_mean`; R^2 uses `cls_token` (each encoder's best aggregation).

4.4 Non-Monotonic Refinement of Vision Encoder Output

Vision encoder provides geometric foundation. Table 3 reports RSA ρ and probe R^2 between representations and actions. DINOv2 `cls_token` achieves the highest RSA ($\rho = 0.692$), exceeding the VLA peak (0.628 at L12) and CLIP (0.557); this ordering is metric-dependent (CKA reverses it; Appendix A.1).

Middle-layer peak (metric-dependent). Under RSA and action-probe decodability, visual representations show a middle-layer peak: trained image_mean ρ is stable through L12 (0.628), then declines to 0.422 at L32 (Figure 2); probe R^2 peaks at L8–L10 (0.655 ± 0.002; Appendix A.15.4). The readout achieves $R^2 = 0.678$ (L13), exceeding the projector by $\Delta = 0.048$ (Table 3). Under linear CKA, this pattern disappears (monotonic increase; Appendix A.1); the non-monotonic trajectory characterizes rank-order geometry specifically, not global covariance structure.

Training primarily enhances readout decodability. Best-vs-best RSA gain is non-significant on LIBERO-Goal ($\Delta\rho = 0.017$, 3%, N=10 tasks; Appendix Table 12), though LIBERO-Spatial yields a significant geometric gain ($\Delta = +0.023$, $p < 0.001$; Appendix A.14). The $N = 10$ instruction set limits statistical power for detecting small effects; non-significant differences should not be interpreted as evidence of absence. Functional improvements are larger: readout R^2 $\Delta = +0.069$ ($p < 0.0001$, 95% CI $[0.047, 0.090]$), image_mean $\Delta = +0.015$ ($p = 0.006$), surviving FDR correction at 29–30/33 layers.

5 Discussion

Synthesis. VLA fine-tuning primarily reorganizes instruction routing rather than visual geometry: training effects concentrate at the readout (R^2 +0.069, attention +46pp) rather than images ($\Delta \leq 1.7\%$). This explains the visual dominance of Grant et al. [3]: DINOv2 has the highest geometric alignment ($\rho = 0.692$), yet VLA readout surpasses it in decodability ($R^2 = 0.678$ vs. 0.634). Seven complementary analyses support instruction-conditioned action generation through distributed L1 routing, where instruction-attention fraction predicts per-head contribution magnitude. *Interventional*: (1) dose-response (8 conditions, $d = 0.48 \rightarrow 0.64$; Figure 8); (2) counterfactual analysis (4 perturbations, $d = 0.63$–0.97; Table 6). *Supporting*: (3) attention gradient ($0 \rightarrow 9 \rightarrow 24$ heads); (4) R^2 ablation gradient (+0.059 at L1); (5) trajectory divergence ($d = 0.65$). *Constraining*: (6) per-head bootstrap CIs (collective d: 0.65 vs. 0.41; $p = 0.0018$); (7) specificity control (1.08×; Table 22).

Mechanistic interpretation. Matched-size ablation (d: instruction 0.65, random 0.60, image 0.58; Table 22) and dose-response (steep through 12 heads, plateau beyond 20) indicate distributed routing; counterfactual ordering follows token novelty ($\rho = 1.0$; §4.3). That ablation raises R^2 yet

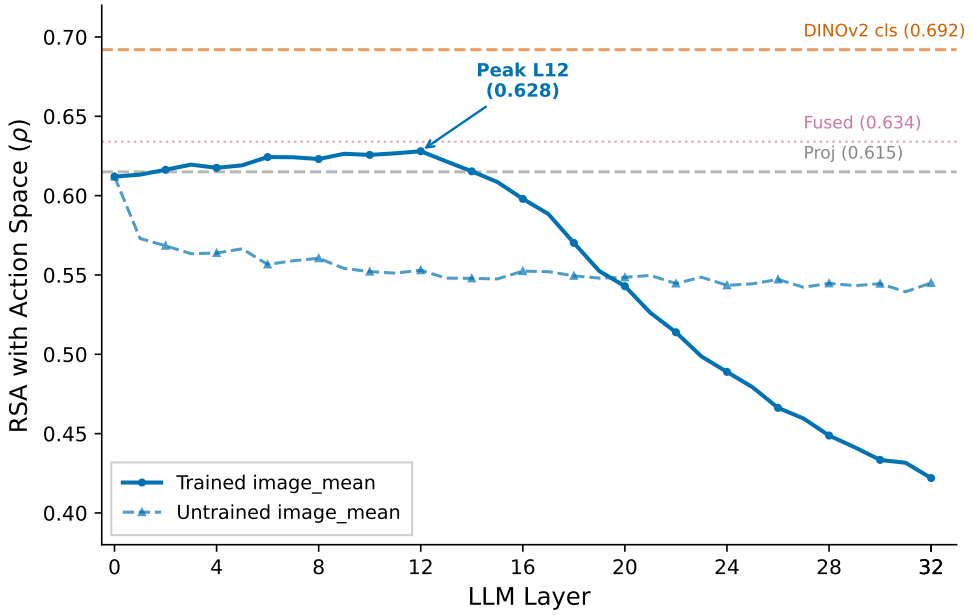


Figure 2: Per-layer RSA (ρ). Trained image_mean follows an inverted-U (L0 $0.612 \rightarrow$ L12 $0.628 \rightarrow$ L32 0.422); untrained decays monotonically. Dashed: DINOv2 cls (0.692), DinoSigLIP (0.634), Projector (0.615).

alters trajectories ($d = 0.65$) without increasing ground-truth error (ΔMSE ≈ 0; Appendix A.8) reflects an information-computation dissociation [42]: routing controls direction rather than precision, and random ablation yields a larger R^2 gain ($+0.077$ vs. $+0.042$) because unstructured perturbation releases probe capacity without disrupting organized computation.

Functional stages and predictions. The per-layer profiles suggest three functional stages: routing (L0–L1; probe 99.6%, 24/32 instruction-dominant heads), computation (L8–L13; R^2 peak 0.655–0.678), and decay (L14–L32, $\sim 56\%$ of parameters; both R^2 and RSA decline, suggesting pruning potential). The entry-exit pattern characterizes general VLA training; task-specific fine-tuning eliminates L31 re-engagement ($18 \rightarrow 0$ heads), suggesting exit routing serves multi-task flexibility (§4.3). Predictions: cross-attention VLAs (π_0 [12], Octo [47]) should lack L1 routing heads; decoder-only VLAs (RT-2 [1]) should show similar routing; larger VLAs should yield flatter per-head contributions.

Task-dependent routing. The routing effect is task-dependent: instruction-discriminative tasks show $2.5\times$ larger $|\Delta R^2|$ under L1 ablation than visually-sufficient tasks ($p = 0.033$; Table 21), rising to $6.5\times$ ($p = 0.008$) under a stricter criterion; no single task drives the effect (Appendix A.22). For deployments, L1 routing probes can flag instruction-discriminative tasks and L1 attention entropy can monitor instruction under-attendance at runtime.

Limitations and scope. We study OpenVLA-7B, trained on multi-platform data (OXE, 970K episodes) and evaluated on held-out LIBERO; the framework generalizes via shared architectural constraints [1, 12, 47], but routing may differ across scales and paradigms (Appendix A.9, A.1). Llama-2's 9/32 pretrained instruction heads may partly reflect text bias; the $+15$-head delta isolates fine-tuning effects. All analyses are open-loop; our routing analysis (*where*) complements Häon et al.'s [4] (*what* is encoded); closed-loop evaluation combining both is a natural extension.

Conclusion. We find that VLA fine-tuning reorganizes instruction routing to entry-exit layers with L1 as the primary hub, supported by seven complementary analyses. Code and data will be released upon acceptance.

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Use of Generative AI. Generative AI tools (Claude, GPT-4) assisted with code implementation and manuscript editing; all scientific content was produced by the authors.

A Supplementary Material

A.1 CKA Analysis Method

We employ Centered Kernel Alignment (CKA) [39] alongside RSA as a complementary perspective. While RSA captures local geometric alignment via rank-order preservation, CKA measures global representational similarity through kernel-based comparisons.

We compute both *linear CKA*, which uses a linear kernel and is sensitive to second-order statistics, and *RDM kernel CKA*, which operates on the same distance matrices as RSA but aggregates them differently. Using both variants tests whether conclusions about representation structure are robust to the choice of similarity metric.

Linear CKA between high-dimensional representations (4,096-D for LLM layers) and the 7-D action space is subject to a known dimension mismatch: the linear kernel in the low-dimensional space may not capture the same structure as in the high-dimensional space. RDM kernel CKA avoids this issue by operating on distance matrices of equal size regardless of the original feature dimensionality.

A.2 Metric Sensitivity: RSA versus CKA

Linear CKA [39] reverses the RSA ordering: untrained LLM (CKA $= 0.089$) $>$ trained LLM (0.083) $>$ DinoSigLIP+Projector (0.081) $>$ DINOv2 (0.066), partially inverting the self-supervised advantage (Figure 3). RDM kernel CKA partially resolves this discrepancy: trained (0.260) $\approx$ Projector (0.257) $>$ untrained (0.246) $>$ DINOv2 (0.241). This divergence reflects the known sensitivity of representation similarity metrics to different geometric properties.

At the per-layer level, linear CKA between image_mean and the action space rises monotonically from L0 (0.081) to L32 (0.098), with no inverted-U pattern. RSA captures rank-order geometric alignment; CKA captures global linear structure. The dissociation indicates that the geometric refinement in early layers operates on the relative arrangement of representations (affecting RSA) without altering the overall covariance structure (captured by CKA).

A.3 Branch-Split Detailed Analysis

To disentangle the contributions of the two encoder streams within DinoSigLIP, we extract representations from the DINOv2 branch (1,024-D) and SigLIP branch (1,152-D) separately before fusion. The two branches yield near-identical RSA: DINOv2 branch $\rho = 0.619$, SigLIP branch $\rho = 0.621$. Fusion produces a representation ($\rho = 0.634$) that exceeds either branch alone; concatenation captures complementary structure. The MLP projector then compresses this signal: DinoSigLIP+Projector achieves $\rho = 0.615$, a drop of $\Delta = 0.019$ from the fused 2,176-D input.

A.4 RSA Across Representation Sources

Figure 4 shows RSA with the action space across all representation sources; the ordering matches Table 3.

A.5 Per-Layer RSA Profiles

Figure 5 extends Figure 2 (main text) by adding readout and untrained profiles.

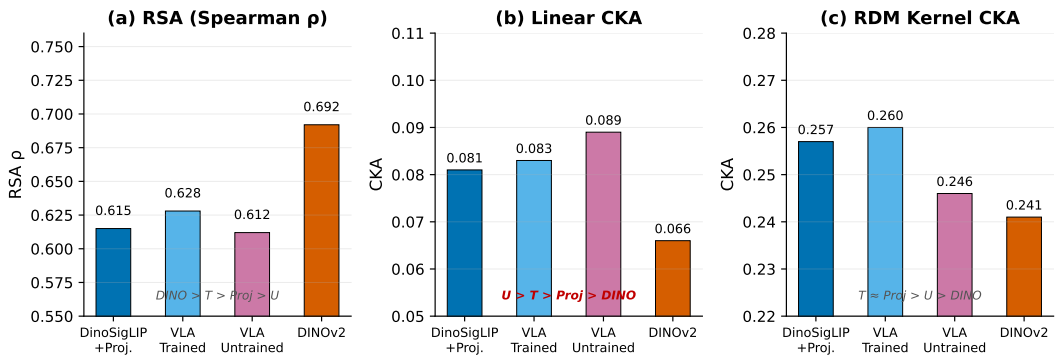


Figure 3: Three similarity metrics across representation sources. **(a)** RSA ranks DINOv2 cls (0.692) > Fused (0.634) > Projector (0.615). **(b)** Linear CKA reverses the ordering: Untrained (0.089) > Trained (0.083) > Projector (0.081) > DINOv2 (0.066). **(c)** RDM kernel CKA: Trained (0.260) ≈ Projector (0.257) > Untrained (0.246) > DINOv2 (0.241). The metric divergence underscores that RSA and CKA capture different aspects of representational structure.

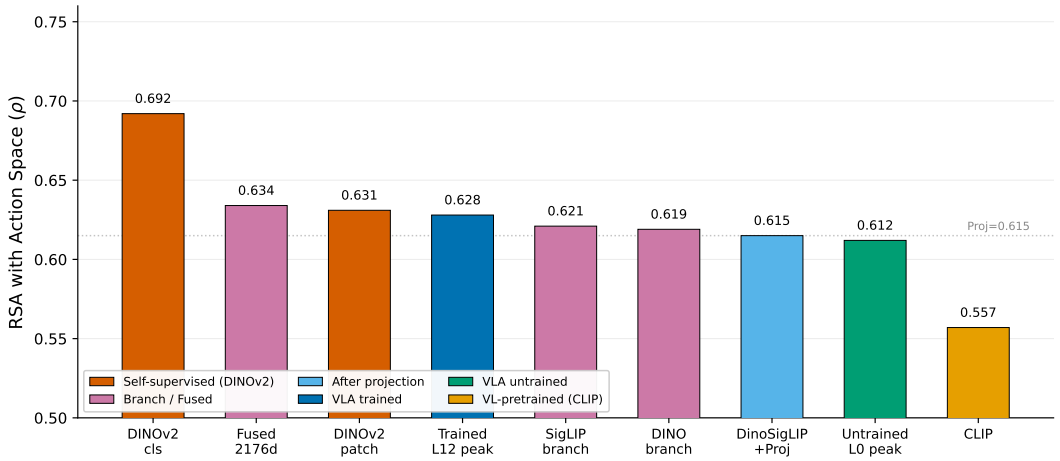


Figure 4: RSA with the action space (ρ) across representation sources. Self-supervised DINOv2 `cls_token` (ρ=0.692) achieves the highest geometric alignment. CLIP (0.557) yields the weakest alignment among pretrained encoders. Error bars: 95% bootstrap CIs.

A.6 Per-Layer Action Decoding Across Training Stages

Figure 6 compares per-layer action probe R^2 across three training stages (untrained, Llama-2 base, VLA-trained) using 50 demonstrations per task, averaged over 3 seeds (mean $\pm$ std). All three conditions share L0 representations ($R^2 = 0.645 \pm 0.008$); subsequent differences thus arise from Transformer processing rather than input variation. The trained model peaks at L9 ($R^2 = 0.643 \pm 0.014$), consistent with progressive specialization for action-relevant feature integration. Llama-2 base shows a steady decline (L0: 0.645 $\rightarrow$ L32: 0.551, std 0.003–0.035), consistent with stable but unspecialized processing. The untrained model exhibits high cross-seed variance and deep-layer collapse, with negative R^2 and bimodal behavior across seeds at late layers; this instability indicates that randomly initialized Transformer layers produce destructive interference on action-relevant structure. The three-condition gradient (trained > Llama-2 base > untrained) holds from L1 through L15, with a reversal at L32 where trained R^2 drops below the pretrained baseline. The qualitative pattern is consistent across all 3 seeds.

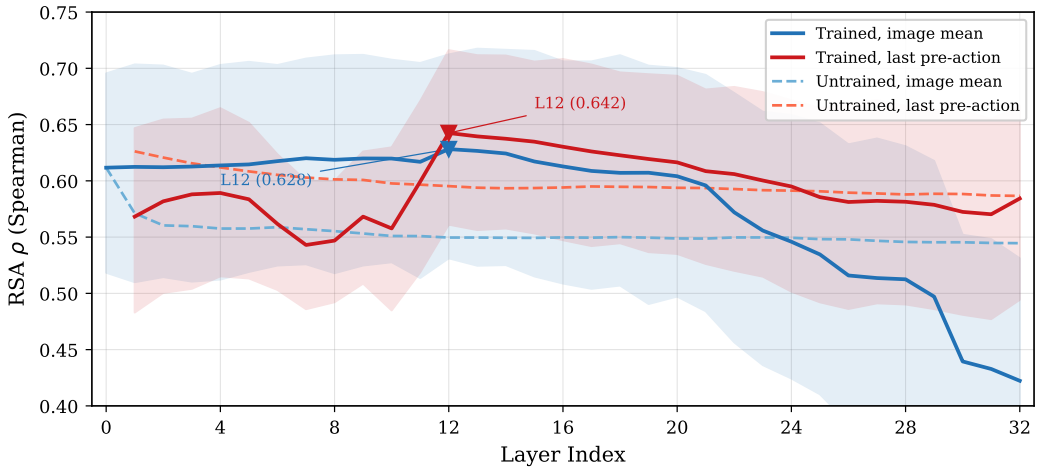


Figure 5: Per-layer RSA (ρ) for trained and untrained models at image_mean and readout (last_preaction) positions. The trained model shows an inverted-U peaking at L12 for both positions; the untrained model shows monotonic decline. After L12, readout and image_mean diverge: readout ρ declines gradually ($0.628 \rightarrow 0.584$) while image_mean ρ drops sharply ($0.628 \rightarrow 0.422$), consistent with task-specific conditioning at the readout position.

A.7 Random-Subset Ablation Details

To evaluate whether the ablation effect in §4.3 is specific to instruction-dominant heads, we compare dominant-head ablation against 10 random subsets of 24/32 L1 heads (each subset drawn without replacement). Dominant-head ablation increases action R^2 by $+0.042$ (within-experiment baseline 0.517), while random subsets yield a mean increase of $+0.077 \pm 0.013$ (Figure 7). The dominant-head effect falls below the random distribution ($z = -2.73$, $p = 0.003$); heads with higher instruction-attention fraction carry more organized routing signals, consistent with graded rather than binary specialization.

A.8 Action Trajectory Analysis Details

Table 7 reports per-dimension mean absolute shift for instruction-dominant (24 heads) and control (8 heads) L1 ablations. Both conditions produce negligible shifts on translational dimensions (dx, dy, $dz < 0.01$) and small rotational shifts (rx, ry, $rz < 0.04$). The gripper dimension absorbs the majority of the effect: instruction-head ablation shifts gripper predictions by 0.291 (control: 0.141), consistent with L1 attention influencing discrete open/close decisions.

Per-task Cohen's d values for instruction-head ablation are 0.43, 0.69, 0.70, 0.70, and 0.73 (mean 0.65), with all tasks significant ($p < 10^{-10}$). Token-level KL divergence between original and ablated action distributions is 5.08 for instruction heads and 1.69 for control heads.

Trajectory-level aggregation. Aggregating per-demo mean actions to the trajectory level reveals that token-level d values are conservative: within-trajectory averaging reduces noise, yielding $\sim 4\times$ larger effect sizes (mean token-to-trajectory ratio 0.245). Instruction-head ablation ($N = 50$: 5 tasks $\times$ 10 demos) yields trajectory $d = 2.943$ (95% CI $[2.469, 3.707]$); bottom-24 non-instruction heads ($N = 30$) yield $d = 2.347$ (CI $[1.854, 3.221]$). All 14 conditions produce trajectory $d > 2.0$ with CI lower bounds > 1.7. Counterfactual trajectory-level d: empty 3.438 (CI $[2.921, 4.448]$), wrong 3.798 ($[3.194, 4.938]$), paraphrased 2.131 ($[1.740, 2.819]$), shuffled 2.595 ($[2.028, 3.791]$). Fine-grained ordering is not reliably distinguished at trajectory level ($N = 30$); token-level comparisons ($N \sim 1{,}500$) provide higher resolution for ordering claims.

Figure 6: Per-layer action probe R^2 (image_mean) across three training conditions: untrained VLA (random weights), Llama-2 base (pretrained, no VLA fine-tuning), and VLA-trained. All conditions share L0 (0.645 ± 0.008). Trained model peaks at L9; Llama-2 base declines steadily; untrained shows high cross-seed variance and deep-layer collapse. 50 demonstrations per task, 3 seeds (mean $\pm$ std).

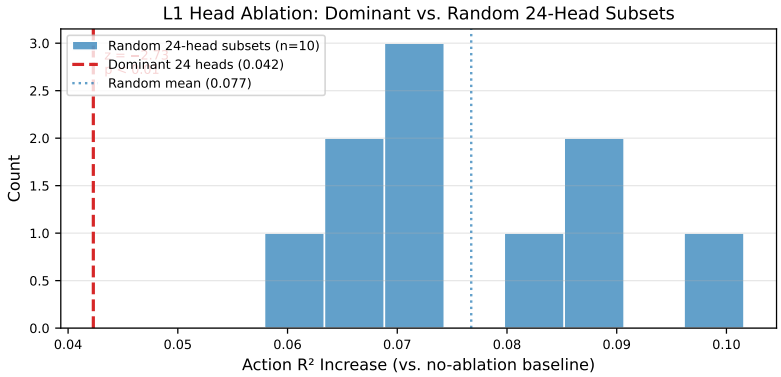


Figure 7: R^2 increase from ablating 24/32 L1 heads. Gray bars: 10 random subsets; red line: dominant-24 heads ($+0.042$). Random subsets yield larger gains (mean $+0.077 \pm 0.013$; $z = -2.73$, $p = 0.003$), consistent with graded routing specialization across L1 heads.

Ground-truth action error. Ablation redirects model outputs ($d = 0.65$) but does not systematically increase error relative to expert demonstrations: original MSE $= 0.498$; top-24 instruction-head ablation MSE $= 0.497$ ($\Delta = -0.001$, Wilcoxon $p = 0.52$, n.s.); bottom-24 non-instruction heads MSE $= 0.506$ ($\Delta = +0.008$). Changes concentrate on the gripper dimension; translation and rotation dimensions are negligibly affected. Per-task breakdown shows all $|\Delta\text{MSE}| < 3\%$ of baseline with all $p > 0.27$; the near-zero aggregate Δ reflects uniformly small per-task effects rather than cancellation across tasks. This is consistent with the information-computation dissociation: routing heads control instruction-specific direction selection rather than action precision.

Table 4 reports per-head bootstrap CIs (10,000 resamples). Individual control heads produce $\sim 1.9\times$ stronger effects ($d = 0.051$) than instruction heads ($d = 0.027$, $p = 0.0018$), consistent across all

5 tasks. The 24 instruction heads nonetheless produce a larger aggregate perturbation ($d = 0.65$ vs. 0.41), consistent with collective coordination rather than individual strength.

Table 4: Per-head bootstrap analysis of L1 ablation effects. Instruction heads are individually weaker but collectively stronger, consistent with coordinated routing.

Group	N heads	Collective d	Per-head d [95% CI]
Instruction-dominant	24	0.65	0.027 [0.026, 0.029]
Control	8	0.41	0.051 [0.048, 0.055]
Difference	—	—	0.024, $p = 0.0018$

Instruction-head ablation produces more uniform perturbation across action dimensions (CV $= 1.30$ vs. control CV $= 2.74$, permutation $p = 0.016$), consistent with structured, language-conditioned computation rather than diffuse noise.

Table 5 and Figure 8 report the dose-response gradient for instruction-head ablation at L1. Trajectory divergence increases monotonically from $d = 0.482$ (4 heads) to $d = 0.639$ (32 heads), with steep rise through 12 heads (slope $\Delta d/\Delta 4 \approx 0.026$), a plateau at 16–20 heads ($\Delta d = 0.001$), and slow increase at 20–32.

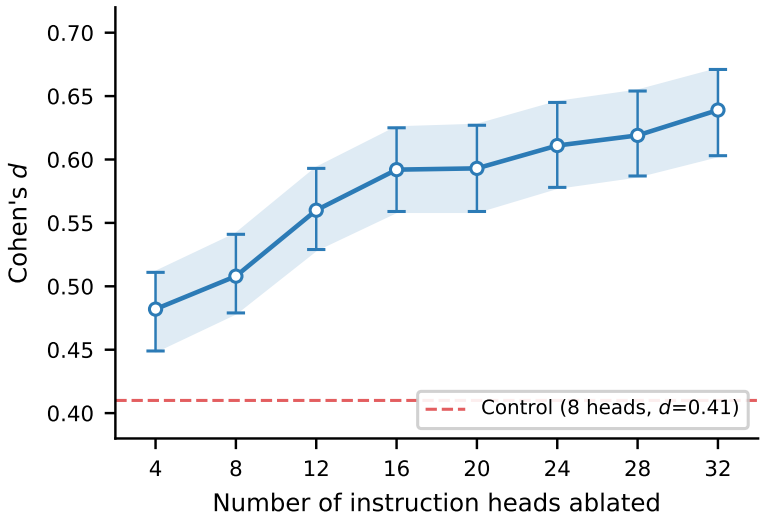


Figure 8: Dose-response gradient: trajectory divergence (Cohen's d) as a function of ablated instruction-dominant heads at L1. Error bars: 95% bootstrap CIs (10,000 resamples). Dashed red: control ablation (8 non-instruction heads, $d = 0.41$). Steep rise through 12 heads, plateau at 16–20, diminishing returns beyond.

Table 5: Dose-response data for L1 instruction-head ablation. Cohen's d vs. original action sequence; 95% bootstrap CIs (10,000 resamples).

Heads ablated	Cohen's d	95% CI
4	0.482	[0.449, 0.511]
8	0.508	[0.479, 0.541]
12	0.560	[0.529, 0.593]
16	0.592	[0.559, 0.625]
20	0.593	[0.559, 0.627]
24	0.611	[0.578, 0.645]
28	0.619	[0.587, 0.654]
32	0.639	[0.603, 0.671]

Table 6: Counterfactual instruction analysis. Cohen's d, token divergence (%), and Jaccard token overlap with original instruction. Token novelty ($1 - \text{Jaccard}$) perfectly predicts divergence ordering (Spearman $\rho = 1.0$, Pearson $r = 0.957$). L1 ablation included for comparison.

Condition	Jaccard	Cohen's d	Token div. (%)
Empty (no instruction)	0.000	0.968	90.7
Wrong (different task)	0.174	0.844	89.5
Paraphrased	0.533	0.697	82.6
Shuffled (random order)	1.000	0.625	85.7
L1 ablation (24 heads)	—	0.650	72.3

Table 7: Mean absolute action shift per dimension under L1 head ablation. Instruction: 24 dominant heads; Control: 8 remaining heads. Gripper shift dominates both conditions.

Condition	dx	dy	dz	rx	ry	rz	Gripper
Instruction (24)	0.002	0.005	0.008	0.021	0.031	0.034	0.291
Control (8)	0.000	0.001	0.002	0.008	0.014	0.018	0.141

A.9 Pretrained LLM Representation Quality Under Visual Input

A potential concern with the three-condition comparison (§3.1) is that the pretrained Llama-2 backbone never encountered projected visual tokens during pretraining, so its representations of such input could be degenerate. We evaluate three diagnostics to assess representation quality. Activation magnitude grows from 16.5 (L0) to 90.2 (L31) for Llama-2 base, compared to $16.5 \rightarrow 51.7$ for the VLA-trained model; the pretrained LLM produces larger but non-exploding activations with no evidence of collapse. Per-layer cosine similarity between L0 and deeper layers decreases progressively from 1.0 to ≈ 0.28 at L25, consistent with gradual differentiation rather than catastrophic out-of-distribution behavior. Top-10 principal component variance explained remains stable across all layers (0.59–0.74), with no low-rank collapse. These diagnostics indicate that the pretrained LLM produces structured, full-rank representations of visual input despite never encountering projected visual tokens during pretraining (Figure 9). These results support the validity of the three-condition comparison.

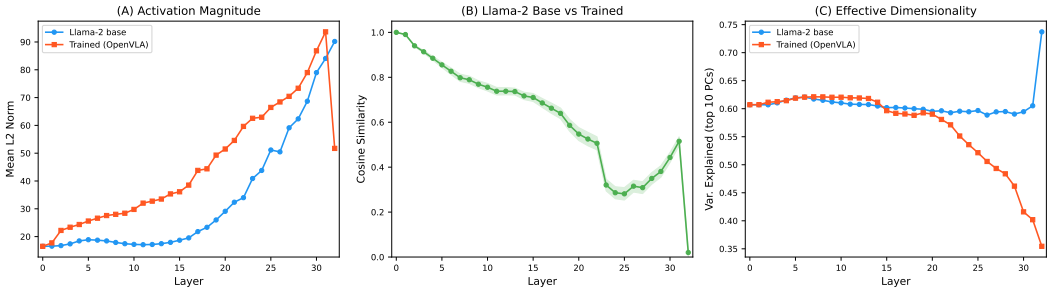


Figure 9: Pretrained LLM representation diagnostics under visual input. Left: activation magnitude across layers (Llama-2 base vs. VLA-trained). Center: cosine similarity between L0 and deeper layers. Right: top-10 PC variance explained. All three metrics indicate structured, non-degenerate representations.

A.10 Probe Architecture Sensitivity

To rule out that our findings depend on MLP probe capacity ($N/\text{params} \approx 0.06$, overparameterized), we replicate the per-layer analysis with a linear probe ($4{,}096 \rightarrow 7$, $N/\text{params} \approx 2.2$). The inverted-U pattern reproduces: trained image_mean peaks at L14 ($R^2 = 0.565$), while untrained declines monotonically from L0 ($R^2 = 0.517$). The trained–untrained gap is larger under the linear

probe (L14: $\Delta = +0.324$) than the MLP probe (L8: $\Delta = +0.015$); VLA training thus reorganizes representations to make action information more linearly decodable. Untrained deep layers collapse entirely under the linear probe (L25: $R^2 = -0.298$); MLP nonlinearity partially rescues these degraded representations. For last_preaction tokens, the linear probe also corroborates the inverted-U (peak L13, $R^2 = 0.596$), with the same peak layer as the MLP probe and a trained-vs-untrained gap of $\Delta = 0.103$ ($p < 0.001$). These results indicate that our qualitative findings reflect representational structure rather than probe capacity artifacts (Figure 10).

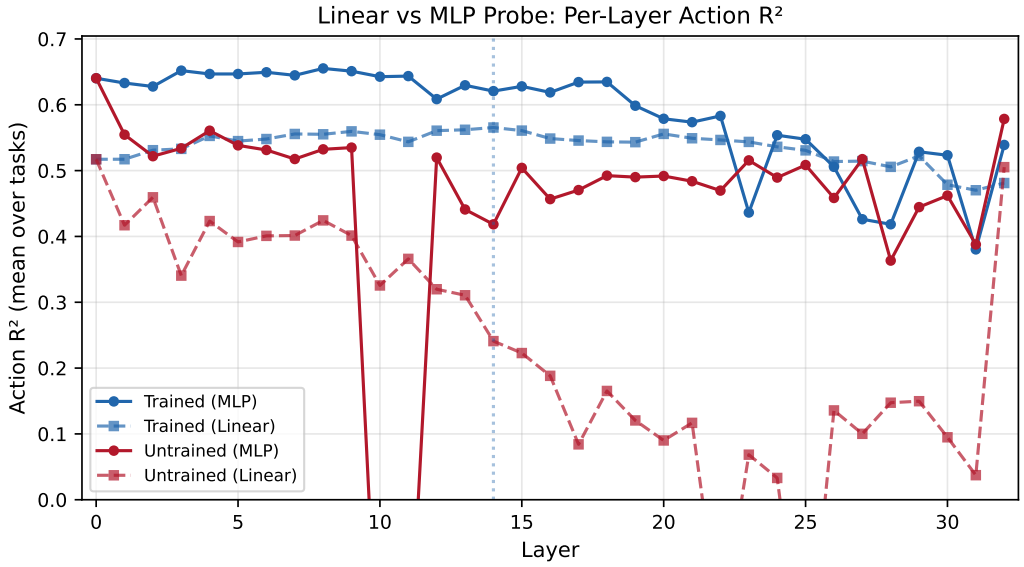


Figure 10: Per-layer action probe R^2 for linear (dashed) and MLP (solid) probes across trained (blue) and untrained (red) conditions. The inverted-U reproduces under both probe architectures; the trained–untrained gap is larger for linear probes.

A.11 Functional Probe Comparison

Figure 11 compares functional probe R^2 across representation sources at each source's best layer.

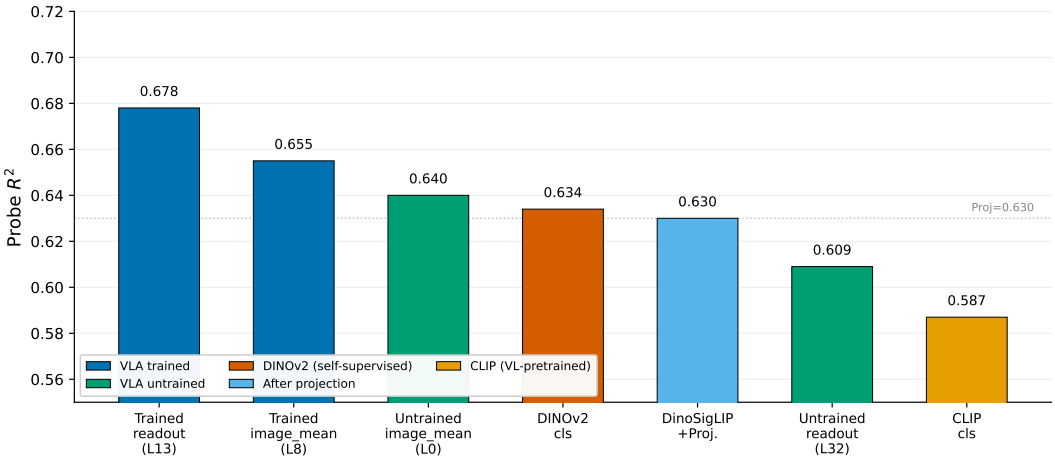


Figure 11: Functional probe R^2 for action prediction (best layer per source). Trained readout (R^2=0.678, L13) and trained image_mean (0.655, L8) exceed the projector baseline (0.630); the inverted-U extends to functional capacity.

A.12 Training Effect and Instruction Invariance

Figure 12 shows trained vs. untrained RSA profiles and instruction invariance. Instruction invariance holds across all conditions: maximum $|\Delta| < 0.006$, so action geometry is visually grounded and not modulated by language instructions.

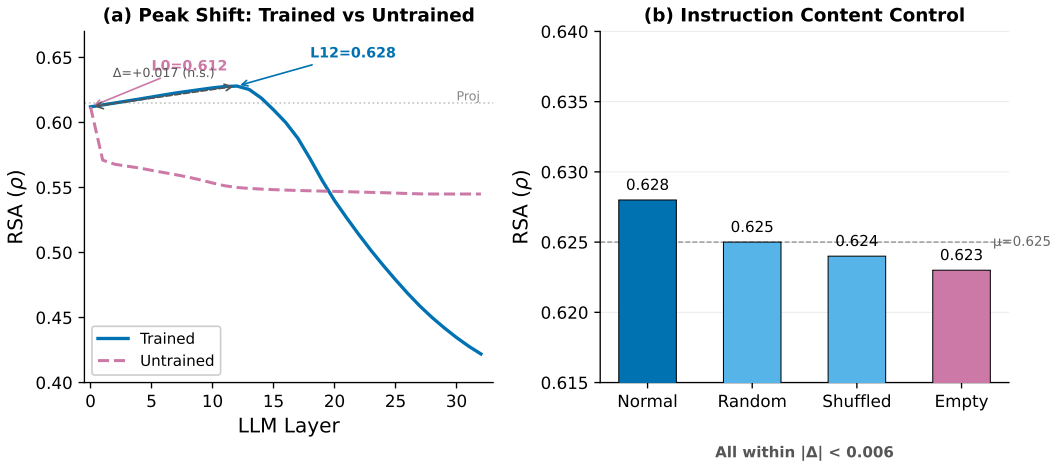


Figure 12: **(a)** Trained vs. untrained image_mean RSA profiles. Training shifts the peak from L0 (0.612) to L12 (0.628), yielding a modest best-vs-best gain ($\Delta = +0.017$, 3%). **(b)** Instruction invariance: all conditions within $|\Delta| < 0.006$; action geometry is visually grounded.

A.13 Additional RSA Values

DINOv2 `patch_mean` achieves RSA $\rho = 0.631$, between the `cls_token` (0.692) and the fused DinoSigLIP (0.634). VLA untrained readout peaks at L1 ($\rho = 0.626$).

A.14 Cross-Benchmark Validation: LIBERO-Spatial

To assess generalization beyond in-distribution tasks, we repeat the core RSA and probe analyses on LIBERO-Spatial (10 out-of-distribution tasks requiring spatial reasoning). Figure 13 shows per-layer RSA and R^2 profiles for both LIBERO-Goal and LIBERO-Spatial side by side.

The inverted-U profile replicates on LIBERO-Spatial: trained image_mean RSA peaks at L14 ($\rho = 0.691$) and R^2 at L13 (0.696), with peaks shifting $\sim$2 layers deeper than LIBERO-Goal (RSA L12, R^2 L8–L10), consistent with increased task difficulty requiring deeper processing. Training produces a significant geometric gain (best-vs-best $\Delta = +0.023$, 95% CI $[0.012, 0.036]$, $p < 0.001$), larger than the LIBERO-Goal effect ($\Delta = +0.017$, CI includes zero).

Two notable differences emerge. First, the absolute RSA values are higher on Spatial (0.691 vs. 0.628), possibly reflecting greater action diversity in spatial tasks that produces a more discriminative action RDM. Second, RSA–R^2 peak convergence does not replicate: readout RSA peaks at L15 while R^2 peaks at L21 (6-layer gap), contrasting with the near-alignment on Goal ($|\Delta_{\text{layer}}| = 1$). This suggests that peak convergence is benchmark-specific rather than a general consequence of training.

A.14.1 Cross-Instruction Probe on LIBERO-Spatial

Table 8 reports cross-instruction 10-way classification accuracy on LIBERO-Spatial. The core pattern from LIBERO-Goal replicates: image_mean remains at chance ($\sim$10%) across all layers and conditions, while the trained readout token saturates to 100% by L4. The untrained readout decays more steeply than on LIBERO-Goal (75.4% at L31 vs. 82%), consistent with greater task difficulty.

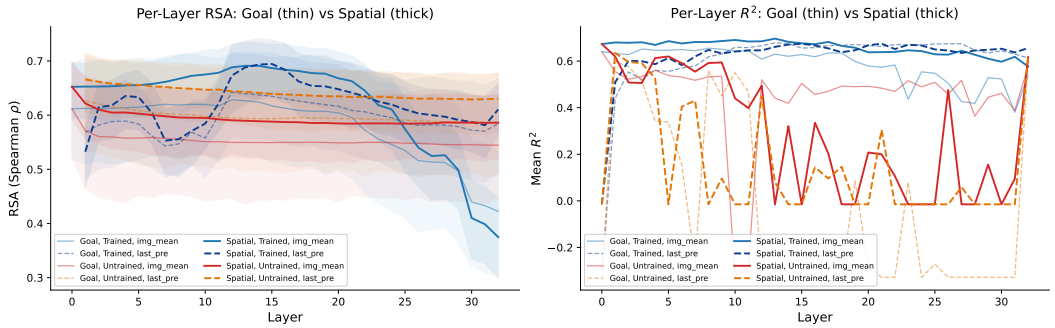


Figure 13: Cross-benchmark validation: per-layer RSA and R^2 profiles on LIBERO-Goal (left) vs. LIBERO-Spatial (right). The inverted-U replicates on both benchmarks. LIBERO-Spatial shows a significant training effect (RSA best-vs-best $\Delta = +0.023$, $p < 0.001$) and deeper peak layers (RSA L14, R^2 L13), but RSA–R^2 peak convergence does not replicate (6-layer gap).

Table 8: Cross-instruction 10-way classification accuracy on LIBERO-Spatial. Trained readout achieves 100% from L4; image_mean remains at chance. Untrained readout decays more steeply than on LIBERO-Goal (cf. Table 1).

Token	Condition	L1	L4	L8	L16	L31
last_preaction	Trained	98.9%	100.0%	100.0%	100.0%	100.0%
last_preaction	Untrained	99.5%	98.3%	85.7%	68.3%	75.4%
image_mean	Trained	10.2%	10.0%	9.9%	10.0%	10.1%
image_mean	Untrained	10.1%	10.0%	10.0%	10.0%	10.2%

A.15 Robustness Analyses

A.15.1 Distance Metric Invariance

The inverted-U profile is robust to distance metric choice: cosine action distance yields identical peak layers (L12) and Pearson correlation $r > 0.989$ with the Euclidean profile reported in the main text (Table 9).

Table 9: Peak layer comparison between Euclidean and cosine action distances. Profile r denotes Pearson correlation across all 32 layers.

Condition	Eucl. Peak	Cos. Peak	Profile r
Trained image_mean	L12	L12	0.990
Trained last_preaction	L12	L13	0.990
Untrained image_mean	L0	L0	0.995
Untrained last_preaction	L1	L1	0.992

A.15.2 Per-DOF Decomposition

RSA decomposition by action dimension indicates the inverted-U is not driven by any single degree of freedom: xyz translation (peak L13, $\rho = 0.193$), rotation (peak L12, $\rho = 0.175$), and gripper (peak L13, $\rho = 0.106$) all exhibit non-monotonic profiles. Per-DOF RSA values are lower than the 7-D RSA because single-dimension distances carry less discriminative information than the full action space.

A.15.3 Permutation Test

Permutation testing (1,000 shuffles) indicates all observed RSA values are highly significant: observed ρ (0.42–0.63) exceeds the null distribution maximum (0.08–0.10) by 15–23 standard deviations (all $p < 0.001$).

A.15.4 Probe Seed Sensitivity

Action probe R^2 is highly stable across random seeds. Over 5 seeds (trained image_mean, full-layer scan), per-layer R^2 standard deviation is uniformly below 0.01 (mean 0.004), and the peak R^2 varies by less than 0.4% (0.655 ± 0.002; Table 10). Peak layer concentrates in the L8–L10 region ($4/5$ seeds); the inverted-U shape is not a seed artifact. Seed 1 peaks at L16, but with R^2 difference <0.002. The inverted-U plateau (L8–L16) is sufficiently flat that argmax is sensitive to minor noise while the profile shape is invariant.

Table 10: Probe R^2 peak across 5 random seeds (trained image_mean).

Seed	Peak Layer	Peak R^2
0	L9	0.654
1	L16	0.656
2	L9	0.653
3	L10	0.655
42	L9	0.655
Mean±std	10.6 ± 2.7	0.655 ± 0.002

A.15.5 Demonstration Subsample Sensitivity

To verify that the inverted-U pattern is not an artifact of sample size, we subsample 25 from 50 demonstrations per task and recompute both RSA and probe profiles (Table 11). The RSA peak layer remains at L12 with a profile correlation of $r = 0.970$ between 50- and 25-demo conditions (high stability). The probe peak shifts from L9 to L11 (within the L8–L16 plateau) with $r = 0.819$; R^2 is more sensitive to training set size than RSA. Both metrics show reduced absolute values (RSA -3.6%, R^2 -7.6%) as expected from halved data, but the non-monotonic profile shape is preserved.

Table 11: Inverted-U stability under demonstration subsampling (trained image_mean).

Metric	50 demos	25 demos	Δ
RSA peak layer	L12	L12	0
RSA peak ρ	0.628	0.606	-3.6%
R^2 peak layer	L9	L11	$+2$
R^2 peak value	0.659	0.609	-7.6%
RSA profile r	—	0.970	—
R^2 profile r	—	0.819	—

A.16 Detailed Training Effects

Table 12 reports bootstrap-based statistical comparisons of training effects on RSA and probe R^2.

A.17 Training Effect Size Summary

Table 13 consolidates training effects across all metrics. Training effects on image_mean metrics are modest (RSA $\Delta = +0.017$, R^2 $\Delta = +0.015$), but effects on last_preaction metrics are substantial (routing $+18$pp, attention $+46$pp, R^2 $\Delta = +0.069$). This asymmetry indicates that training primarily reshapes how the readout token integrates information rather than altering the visual representations carried by image tokens.

A.18 Instruction Routing Probes

A.18.1 Cross-Instruction Classification: Dense Layer Profile

Table 14 shows the dense per-layer profile of cross-instruction classification (10-way) at the readout token for layers L0–L8. The step-function onset between L0 (10%, chance) and L1 ($\geq 99.6\%$) re-

Table 12: Training effects on action-relevant structure (LIBERO-Goal; bootstrap, 10,000 resamples). RSA best-vs-best yields a modest geometric gain ($\Delta = +0.017$, 3%; LIBERO-Spatial: $+0.023$, $p<0.001$), while R^2 shows substantially larger functional gains.

Comparison	Δ	95% CI	p
RSA (ρ)			
Best-vs-best image_mean (trained L12 vs. untrained L0)	+0.017	incl. 0	n.s.
Same-layer L12 image_mean (trained vs. untrained)	+0.079	excl. 0	<0.0001
Instruction effect (max $\lvert\Delta\rvert$)	0.006	—	n.s.
Probe R^2			
Best-vs-best image_mean (trained L8 vs. untrained L0)	+0.015	excl. 0	0.006
Best-vs-best readout (trained L13 vs. untrained L32)	+0.069	excl. 0	<0.0001

Table 13: Training effect sizes across metrics. Image_mean effects are modest; last_preaction effects are large. Training targets readout-token information integration.

Metric	Trained	Untrained	Δ
RSA peak (image_mean)	0.628 (L12)	0.612 (L0)	+0.017
R^2 peak (image_mean)	0.655 (L8)	0.640 (L0)	+0.015
R^2 peak (last_preaction)	0.678 (L13)	0.609 (L32)	+0.069
Instruction routing L31	100%	82%	+18pp
Instruction attention L1	58%	12%	+46pp
Progress classification	86.3% (L20)	85.3% (L4)	+1pp

flects the high discriminability of instruction embeddings from the frozen tokenizer: even incidental attention ($\sim$12%) suffices for linear classification at L1. Both pretrained and VLA-trained models sustain 100% through L31; only random initialization shows progressive decay (82% at L31). Deep-layer stability thus derives from structured pretrained weights rather than task-specific fine-tuning.

Table 14: Cross-instruction 10-way classification accuracy at readout token, L0–L8. Values reflect dense routing evaluation; main-text Table 1 reports sparse single-split results, yielding minor differences (e.g., L8: 98.2% vs. 98.6%).

Layer	Trained	Untrained	Image_mean (trained)
L0	10.0%	10.0%	10.0%
L1	100.0%	100.0%	10.0%
L2	100.0%	99.6%	10.0%
L3	100.0%	99.6%	10.0%
L4	100.0%	99.1%	10.0%
L5	100.0%	98.2%	10.0%
L6	100.0%	97.3%	10.0%
L7	100.0%	96.4%	10.0%
L8	100.0%	98.6%	10.0%

A.18.2 Cross-Instruction LOOCV Validation

Leave-one-out cross-validation over 10 folds (each fold holds out one task's instructions) validates that the cross-instruction probe result is not driven by memorization. Trained last_preaction achieves $100\% \pm 0.00$ across all 10 folds at layers L1–L31 (perfect generalization).

A.18.3 QA Probe: Progress and Subtask Classification

To test whether LLM layers encode task-relevant semantic information beyond instruction identity, we train probes to classify task progress (early/mid/late) and active subtask from frozen representations (Table 15).

Table 15: QA probe accuracy (progress + subtask classification, best layer).

Source	Best Layer	Accuracy
Trained last_preaction	L20	86.3%
Untrained last_preaction	L20	85.3%
DINOv2 cls_token	—	82.7%

A.18.4 Cross-Instruction Probe Profiles

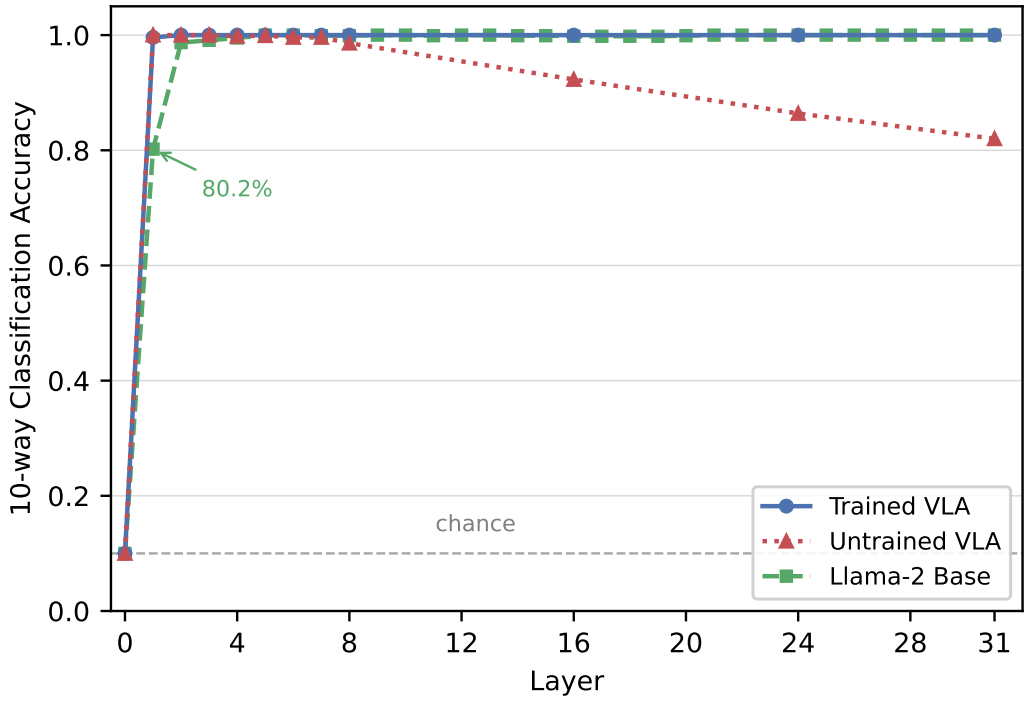


Figure 14: Cross-instruction 10-way classification accuracy at the readout token across all 32 LLM layers. VLA-trained achieves 99.6% at L1, saturating to 100%. Llama-2 base starts at 80.2%, reaching 100% by L5. Untrained starts at 100% but decays to 82% by L31.

A.18.5 L1 Attention Pattern Analysis

Figure 15 shows the per-head L1 attention distribution at the readout token. The three-level gradient (untrained 0/32, Llama-2 base 9/32, VLA-trained 24/32 instruction-dominant heads) shows that language pretraining seeds partial instruction routing and VLA fine-tuning shifts it further toward instruction dominance.

A.18.6 Multi-Layer Attention Patterns

Table 16 shows a 4-layer subset of the full 32-layer attention profile (Figure 1, main text). The full profile (Figure 16) reveals three qualitatively distinct patterns: the untrained model maintains 0 instruction-dominant heads at all 32 layers; Llama-2 base shows a mid-depth peak (L9–L16: 15–22

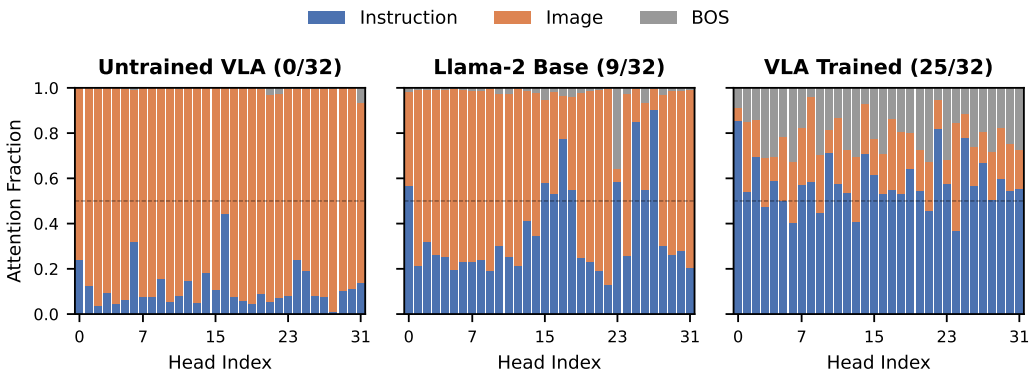


Figure 15: L1 attention from the readout token across three conditions, per head (sorted by instruction attention fraction). VLA-trained: 24/32 instruction-dominant heads (57.7% text attention). Llama-2 base: 9/32 heads (36.5%). Untrained: 0/32 heads (11.7%). See Table 2 for quantitative comparison.

heads), consistent with typical autoregressive LM processing depth; VLA fine-tuning produces an entry-exit pattern (L0–L1: 21–24 heads, L31: 18 heads, intermediate layers 0–11).

Table 16: Instruction-dominant heads and text attention fraction at four sampled layers (subset of the 32-layer profile in Figure 1). VLA-trained shows entry-exit concentration (L1, L31); Llama-2 base peaks at mid-depth (L16); untrained shows no instruction routing. Head counts from the 32-layer bulk extraction may differ slightly from dedicated per-layer analyses (Table 2) due to threshold boundary effects.

	VLA Trained		Llama-2 base		Untrained	
Layer	Heads	Text%	Heads	Text%	Heads	Text%
L1	**24/32**	57.7	9/32	36.5	0/32	11.7
L8	4/32	46.8	8/32	48.8	0/32	7.9
L16	10/32	48.4	**25/32**	60.2	0/32	9.2
L31	**18/32**	56.3	8/32	34.9	0/32	7.6

A.18.7 Threshold Sensitivity Analysis

The main text defines instruction-dominant heads as those allocating $> 50\%$ attention to instruction tokens. Table 17 shows that the three-condition gradient (untrained $<$ Llama-2 base $<$ VLA-trained) holds across all thresholds from 30% to 70%; the 50% threshold is not a boundary-sensitive choice.

Table 17: Number of instruction-dominant heads at L1 under varying attention thresholds. The three-condition ordering is stable across all thresholds.

Threshold	VLA Trained	Llama-2 base	Untrained
30%	32	13	0
40%	31	10	0
50%	24	9	0
60%	9	4	0
70%	5	3	0

A.19 L1 Attention Decomposition by Head Group

Table 18 decomposes L1 attention mass at the readout token into three token categories for instruction-dominant and control head groups.

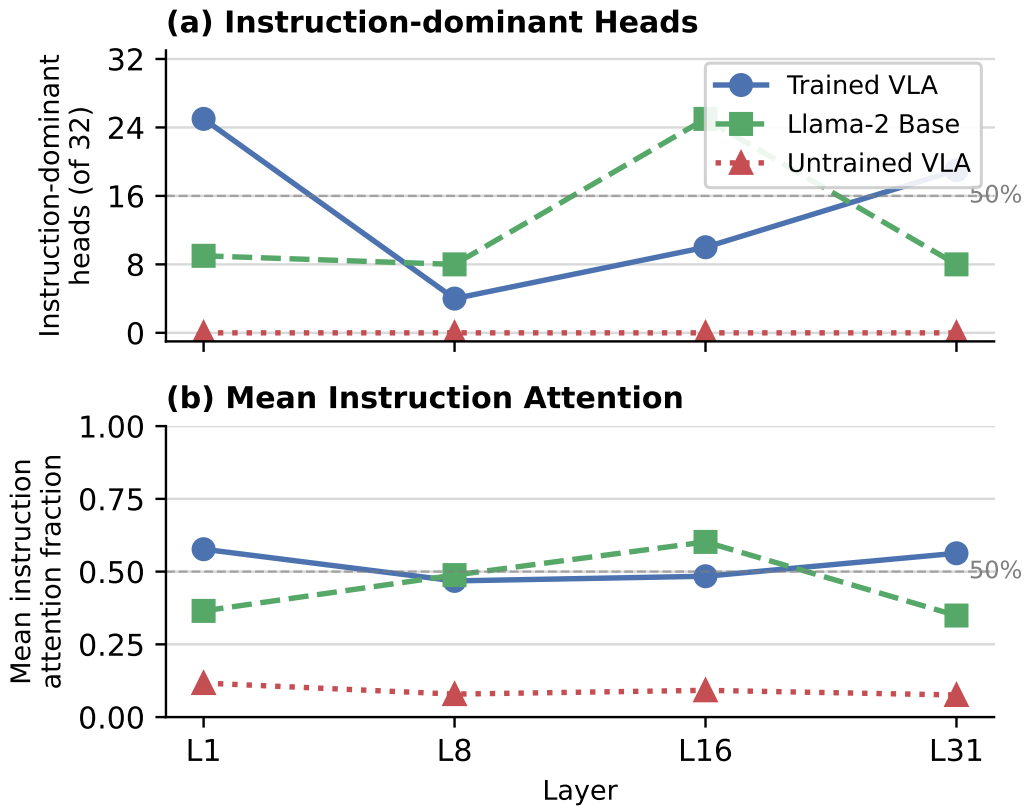


Figure 16: Instruction-dominant heads and mean text attention fraction across layers for three conditions. VLA-trained shows elevated values at L1 and L31; Llama-2 base peaks at L16; untrained remains near zero throughout.

Table 18: Attention mass allocation at Layer 1, averaged across 5 samples. Instruction-dominant heads (defined as $> 50\%$ attention to instruction tokens) allocate 17.5 percentage points more attention to instruction tokens than control heads, with corresponding reductions in vision (-8.3pp) and BOS/sink (-9.2pp) attention.

Head Group	N	Instruction	Vision	BOS/Other
Instruction-dominant	24	62.4%	18.8%	18.8%
Control	8	44.9%	27.1%	28.0%
Δ	—	+17.5pp	−8.3pp	−9.2pp

Head-level instruction attention entropy reaches 99.5% of the uniform maximum; routing is broadly distributed rather than concentrated in a few heads.

Per-token attention normalization. The 57.7% aggregate text attention in instruction-dominant heads partly reflects token-count imbalance (22 instruction vs. 256 image tokens). Per-token normalization reveals a $38.2\times$ ratio: each instruction token receives $38.2\times$ the attention of each image token in trained instruction-dominant heads, compared to only $1.5\times$ in untrained heads (a $25.5\times$ increase from training). Instruction-dominant heads also show lower attention entropy (0.465) than non-dominant heads (0.569), indicating more concentrated attention distributions.

A.20 Per-Dimension Action MAE Under Head Ablation

Table 19 extends Table 7 with full per-dimension control-head data and instruction-to-control MAE ratios.

Table 19: Mean absolute error per action dimension under L1 head ablation. Translation dimensions (dx/dy/dz) show 3–5× larger instruction-to-control ratio than rotation dimensions (1.9–2.6×); instruction-dominant heads have disproportionate influence on positional control. Gripper (2.07×) falls between the two groups.

Dimension	Instr. 24h MAE	Control 8h MAE	Ratio
dx	0.00189	0.00037	5.09×
dy	0.00531	0.00125	4.26×
dz	0.00762	0.00243	3.14×
$roll$	0.02059	0.00785	2.62×
$pitch$	0.03071	0.01387	2.21×
yaw	0.03388	0.01783	1.90×
$gripper$	0.29072	0.14072	2.07×

A.21 Per-Task R^2 Distribution Under L1 Ablation

Table 20 reports per-task action probe R^2 at L1 under baseline and 24-head ablation conditions (MLP probe, 5 demonstrations per task).

Table 20: Per-task action probe R^2 at L1 (MLP probe, 5 demos/task). All 10 tasks show positive ablation effect ($\Delta > 0$): instruction-head ablation uniformly increases action decodability at the entry layer. Mean $\Delta R^2 = +0.107$; the 50-demo setting (§4.3) yields a consistent $+0.059$.

Task	Baseline R^2	Ablated R^2	ΔR^2
open middle drawer	0.197	0.221	+0.024
open top drawer + bowl	0.265	0.340	+0.075
push plate to stove	0.190	0.197	+0.007
put bowl on plate	0.375	0.474	+0.099
put bowl on stove	0.235	0.395	+0.160
put bowl on cabinet	0.256	0.457	+0.201
put cream cheese in bowl	0.186	0.345	+0.159
put wine on rack	0.189	0.424	+0.235
put wine on cabinet	0.279	0.353	+0.074
turn on stove	0.385	0.421	+0.036
Mean ± std	0.256 ± 0.070	0.363 ± 0.088	+0.107

A.22 Task Complexity and Routing Dependence

We classify tasks as *instruction-discriminative* when multiple tasks share the same manipulated object(s), requiring instruction content to disambiguate the target action (e.g., "put bowl on plate" vs. "put bowl on stove"); remaining tasks involve unique object-action pairings recoverable from visual context alone. Table 21 reports per-task $|\Delta R^2|$ under both groupings.

Leave-one-out sensitivity. All leave-one-out p-values remain ≤ 0.083, with no single task removal causing a qualitative reversal of the effect direction. Reclassifying the boundary case (cream cheese in bowl) to the instruction-discriminative group strengthens the result ($p = 0.008$).

A.23 Specificity Control: Matched-Size Head Ablation

To test whether ablation effects are specific to instruction-dominant heads, we ablate four matched-size subsets of 24 heads at L1: instruction-dominant (highest instruction attention fraction), image-

Table 21: Task-level L1 ablation effects ($|\Delta R^2|$) grouped by instruction dependence. *Primary* (conservative): cream cheese in bowl classified as visual-sufficient ($2.5\times$, $p = 0.033$). *Sensitivity* (shared-object criterion): cream cheese reclassified as instruction-discriminative ($6.5\times$, $p = 0.008$).

Task	$\lvert\Delta R^2\rvert$	Group
put wine on rack	0.235	Instr-disc
put bowl on cabinet	0.201	Instr-disc
put bowl on stove	0.160	Instr-disc
put bowl on plate	0.099	Instr-disc
open top drawer + bowl	0.075	Instr-disc
put wine on cabinet	0.074	Instr-disc
put cream cheese in bowl†	0.159	Vis-suff
turn on stove	0.036	Vis-suff
open middle drawer	0.024	Vis-suff
push plate to stove	0.007	Vis-suff
Primary: Instr-disc mean 0.141; Vis-suff mean 0.057		
Mann-Whitney $U = 21.0$, $p = 0.033$, ratio $= 2.49\times$		
Sensitivity†: cream cheese as Instr-disc: $p = 0.008$, $6.5\times$		

dominant (highest image attention fraction), random (5 seeds), and bottom-24 (lowest instruction attention fraction; overlaps 16/24 heads with the instruction-dominant group).

Table 22: Matched-size ablation controls: trajectory disruption (d) from ablating 24-head subsets at L1, ranked by instruction attention fraction. Instruction-dominant heads produce the largest effect, but the modest gap (8% over random) indicates distributed, attention-graded routing rather than a dedicated subset.

Condition (24 heads)	Cohen's d	95% CI
Instruction-dominant	0.650	(reference)
Random (5 seeds)	0.601 ± 0.021	$[0.568, 0.622]$
Image-dominant	0.580	$[0.547, 0.613]$
Bottom-24	0.565	—

All conditions yield comparable ground-truth MSE (0.487–0.504 vs. baseline 0.497), consistent with direction-selective rather than precision-degrading effects.